

INNER PRODUCT SPACES

Dr. Juhi Kesarwani

Dr. Ashish Kumar Kesarwany

Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University, Bhopal-Indore Highway, Kothrikalan, Sehore, Madhya Pradesh, 466114, India

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Dot Product

Definition (Dot product)

For $x, y \in \mathbb{R}^3$, the dot product of x and y , denoted by $x \cdot y$, is defined by

$$x \cdot y = x_1y_1 + x_2y_2 + x_3y_3,$$

where $\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$.

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where $\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$.

Example

Consider $x = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ then

$$x \cdot y = -2 + 1 + 1 = 0$$

Properties of dot product

The dot product on $\mathbb{R}^3$ has the following properties.

$$* \quad x \cdot y = y \cdot x.$$

$$* \quad x \cdot (y + z) = x \cdot y + x \cdot z.$$

$$* \quad (\alpha x) \cdot y = \alpha(x \cdot y)$$

$$* \quad x \cdot x \geq 0 \text{ and } x \cdot x = 0 \text{ if and only if } x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

for all $x, y \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$.

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Length of a vector

For all $x \in \mathbb{R}^3$, the length of x is denoted by $|x|$ and is defined as

$$|x| = \sqrt{x \cdot x}$$

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$$|x| = \sqrt{x \cdot x}$$

Remark

If we think of a vector as a point instead of as an arrow, then $|x|$ should be interpreted to mean the distance from the origin to the point x .

Angle between two vectors

Let $x, y \in \mathbb{R}^3$ be two non-zero vectors then angle θ between these two vectors is given by

$$\cos \theta = \frac{x \cdot y}{|x||y|}$$

Thus, from here, we get another definition of dot product, i.e.,

$$x \cdot y = |x||y| \cos \theta.$$

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Example

Let $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $x \cdot y = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 = 6$,

$$|x| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \quad \text{and} \quad |y| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos \theta = \frac{x \cdot y}{|x||y|} = \frac{6}{\sqrt{42}} \implies \theta = \cos^{-1} \left(\frac{6}{\sqrt{42}} \right)$$

Inner Product Space

Definition (Inner Product Space)

Let V be a real vector space then an inner product on V is a function that associates a real number $\langle x, y \rangle$ to each pair of vectors x and y in V such a way that the following rules are satisfied for all $x, y, z \in V$ and $\alpha \in \mathbb{R}$:

$$\textcircled{1} \quad \langle x, y \rangle = \langle y, x \rangle.$$

$$\textcircled{2} \quad \langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle.$$

$$\textcircled{3} \quad \langle \alpha x, y \rangle = \alpha \langle x, y \rangle.$$

$$\textcircled{4} \quad \langle x, x \rangle \geq 0 \text{ and } \langle x, x \rangle = 0 \text{ iff } x = 0.$$

A pair $(V, \langle \cdot \cdot \rangle)$ of a vector space V and an inner product $\langle \cdot \cdot \rangle$ is called a **inner product space**. In particular, the pair $(\mathbb{R}^n, \cdot)$ is called the **Euclidean n -space**.

Example

Consider the vector space $\mathbb{R}^3(\mathbb{R})$. For $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ in $\mathbb{R}^3$ define

$$\langle x, y \rangle = x_1y_1 + x_2y_2 + x_3y_3.$$

Then $\langle, \rangle$ is an inner product on $\mathbb{R}^3$.

Example

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$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Then $\langle, \rangle$ is an inner product on $\mathbb{R}^3$.

Example

Let $V = C[0, 1]$ be the vector space of all real-valued continuous functions on $[0, 1]$. For any two functions $f(x)$ and $g(x)$ in V , define

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$$

Then $\langle, \rangle$ is an inner product on V .

Exercise 1:

Let $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$. Show that

$$\langle u, v \rangle = 2u_1v_1 + 3u_2v_2$$

is an inner product on $\mathbb{R}^2$.

Exercise 2:

Consider the vector space $\mathbb{R}_2[x]$. Let

$p(x) = a_0 + a_1x + a_2x^2$, $q(x) = b_0 + b_1x + b_2x^2 \in \mathbb{R}_2[x]$. Show that

$$\langle p(x), q(x) \rangle = a_0b_0 + a_1b_1 + a_2b_2$$

is an inner product on $\mathbb{R}_2[x]$.

Definition(Magnitude or Length or Norm)

Let $(V, \langle \cdot \rangle)$ be an inner product space. By the fourth axiom of an inner product, $\langle u, u \rangle$ is non-negative for any vector $u \in V$. Thus, its positive square root exists. We use the notation

$$\|u\| = \sqrt{\langle u, u \rangle}.$$

This non-negative number is called the **norm** or **length** or **magnitude** of u . The relation $\|u\|^2 = \langle u, u \rangle$ will be used frequently.

The distance between two vectors u and v , denoted by $d(u, v)$, is defined as

$$d(u, v) = \|u - v\|.$$

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The distance between two vectors u and v , denoted by $d(u, v)$, is defined as

$$d(u, v) = \|u - v\|.$$

Remark

If $\|u\| = 1$ or, equivalently, if $\langle u, u \rangle = 1$, then u is called a **unit vector**. Every non-zero vector v in V can be multiplied by the reciprocal of its length to obtain the unit vector

$$\vec{v} = \frac{v}{\|v\|}$$

which is a positive multiple of v . This process is called **normalizing** v .

The angle between two vectors

The real number θ in the interval $[0, \pi]$ that satisfies

$$\langle u, v \rangle = \|u\| \|v\| \cos \theta \quad \text{or} \quad \cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

is called the **angle** between u and v .

Example

In $\mathbb{R}^2$ equipped with an inner product $\left\langle x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\rangle = 2x_1y_1 + 3x_2y_2$, the angle between $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is computed as

$$\begin{aligned}\cos \theta &= \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{2}{\sqrt{14 \times 2}} = \frac{1}{\sqrt{7}} \\ \implies \theta &= \cos^{-1} \left(\frac{1}{\sqrt{7}} \right)\end{aligned}$$

Properties of length in an inner product space V

✿ $\|u\| \geq 0.$

✿ $\|u\| = 0$ if and only if $u = 0.$

✿ $\|ku\| = |k|\|u\|.$

✿ $\|u + v\| \leq \|u\| + \|v\|$

(Triangular inequality)

Exercise 3:

Consider the inner product space $(\mathbb{R}_2[x], \langle \rangle)$ equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx \quad \text{for } f, g \in \mathbb{R}_2[x].$$

Find the angle between $f(x) = 3x - 1$ and $g(x) = x^2$.

Exercise 4:

Consider the inner product space $(C[0, 1], \langle \rangle)$ equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx \quad \text{for } f, g \in C[0, 1].$$

Then

1. Find the angle between 1 and x^2 .
2. Compute the following inner products:

(i) $\langle x, x \rangle$.

(ii) $\langle x, 1 + x^2 \rangle$.

Exercise 5:

Consider the inner product space $(C[0, 1], \langle \cdot, \cdot \rangle)$ equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx \quad \text{for } f, g \in C[0, 1].$$

Then find

(i) $\|f\|$

(ii) $d(f, g)$

(iii) $\langle f, g \rangle$.

Cauchy-Schwarz Inequality

Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality states that for any two vectors u and v in an inner product space $(V, \langle \cdot, \cdot \rangle)$, the absolute value of their inner product is less than or equal to the product of their norms:

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

Proof

Define a function $f(t) = \|tu - v\|^2$, where t is a scalar. Note that $f(t)$ is always non-negative since it is the square of the norm of a vector. Expand $f(t)$ using the definition of inner product and simplify it:

$$f(t) = \|tu - v\|^2$$

$$f(t) = \langle tu - v, tu - v \rangle$$

$$f(t) = t^2 \|u\|^2 - 2t \langle u, v \rangle + \|v\|^2$$

Since $f(t)$ is non-negative, its discriminant must be non-positive:

$$(2 \langle u, v \rangle)^2 - 4 \|u\|^2 \|v\|^2 \leq 0$$

Simplify the discriminant:

$$\langle u, v \rangle^2 - \|u\|^2 \|v\|^2 \leq 0$$

Take the square root of both sides to obtain the Cauchy-Schwarz inequality:

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

This proves the Cauchy-Schwarz inequality.

Definition (Orthogonal Vectors)

Let $(V, \langle \cdot \rangle)$ be an inner product space. Then two vectors u and v of V are said to be **orthogonal** or **perpendicular** if $\langle u, v \rangle = 0$.

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Remark

Note that for non-zero vectors u and v , $\langle u, v \rangle = 0$ if and only if $\theta = \frac{\pi}{2}$.

* The word **orthogonal** comes from the Greek word **orthogonios**, which means right-angled.

Example

Consider the inner product space $(\mathbb{R}^3, \langle \rangle)$ where $\langle \rangle$ is defined as

$$\left\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \right\rangle = x_1x_2 + y_1y_2 + z_1z_2$$

Since $\left\langle \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\rangle = 1 \cdot 1 + 1 \cdot 1 + 2 \cdot (-1) = 0$ and hence vectors $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ are orthogonal vectors.

Exercise 6:

Find a non-zero vector w that is orthogonal to $u_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$ in $\mathbb{R}^3$.

Pythagorean Theorem

Let $(V, \langle \cdot \rangle)$ be an inner product space, and let x and y be any two vectors in V with the angle θ . Then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\| \|y\| \cos \theta.$$

Moreover, it deduces the Pythagorean theorem: $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ for any orthogonal vector x and y .

Proof

Since angle between vectors x and y is θ and therefore

$$\langle x, y \rangle = \|x\| \|y\| \cos \theta$$

By definition

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\ &= \|x\|^2 + \|y\|^2 + 2\|x\| \|y\| \cos \theta\end{aligned}$$

If vectors x and y are orthogonal then by putting $\theta = \frac{\pi}{2}$, we get

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Parallelogram Equality

Let $(V, \langle \rangle)$ be an inner product space. Suppose $u, v \in V$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

Parallelogram Equality

Let $(V, \langle \cdot \rangle)$ be an inner product space. Suppose $u, v \in V$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

Proof

We have

$$\begin{aligned}\|u + v\|^2 + \|u - v\|^2 &= \langle u + v, u + v \rangle + \langle u - v, u - v \rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle + \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle \\ &= 2(\|u\|^2 + \|v\|^2),\end{aligned}$$

as desired.

Exercise 7:

Let $(V, \langle \rangle)$ be an inner product space. Suppose $u, v \in V$ are such that

$$\|u\| = 3, \|u + v\| = 4, \|u - v\| = 6$$

What number must $\|v\|$ equal?

Exercise 8:

Prove that if V is a real inner-product space, then

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

Example

Let $(V, \langle \cdot \rangle)$ be an inner product space then for any two vectors x and y in V

$$|\|x\| - \|y\|| \leq \|x - y\|$$

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$$|\|x\| - \|y\|| \leq \|x - y\|$$

Solution

$$\begin{aligned}\|x\| &= \|(x - y) + y\| \leq \|x - y\| + \|y\| && \text{[Triangula inequallity]} \\ \implies \|x\| - \|y\| &\leq \|x - y\| && (1)\end{aligned}$$

Again,

$$\begin{aligned}\|y\| &= \|(y - x) + x\| \leq \|y - x\| + \|x\| && \text{[Triangula inequallity]} \\ \implies \|y\| - \|x\| &\leq \|y - x\| \\ \implies -(\|x\| - \|y\|) &\leq \|x - y\| && (2)\end{aligned}$$

From (1) and (2), we have

$$|\|x\| - \|y\|| \leq \|x - y\|$$

Example

Let $(V, \langle \rangle)$ be an inner product space and if $x_1, x_2, \dots, x_k$ are non-zero vectors in V that are mutually orthogonal then they are linearly independent.

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Solution

Suppose $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k = 0$. Then for each $i = 1, \dots, k$,

$$\begin{aligned}\langle \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k, x_i \rangle &= \langle 0, x_i \rangle = 0 \\ \implies \alpha_1 \langle x_1, x_i \rangle + \alpha_2 \langle x_2, x_i \rangle + \dots + \alpha_i \langle x_i, x_i \rangle + \dots + \alpha_k \langle x_k, x_i \rangle &= 0 \\ \implies \alpha_i \|x_i\|^2 &= 0 \\ \implies \alpha_i = 0, \quad \forall \quad i = 1, \dots, k\end{aligned}$$

This shows that the vectors $x_1, x_2, \dots, x_k$ are linearly independent.

Matrix representation of inner product

Let $(V, \langle \rangle)$ be an inner product space and let $\{x_1, x_2, \dots, x_m\}$ be an ordered basis of V . Then for any $x = \sum_{i=1}^m \alpha_i x_i$ and $y = \sum_{i=1}^m \beta_i x_i$, we have

Matrix representation of inner product

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$$\begin{aligned}\langle x, y \rangle &= \langle \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_m x_m, \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_m x_m \rangle \\ &= \alpha_1 \beta_1 \langle x_1, x_1 \rangle + \alpha_1 \beta_2 \langle x_1, x_2 \rangle + \cdots + \alpha_1 \beta_m \langle x_1, x_m \rangle \\ &\quad + \alpha_2 \beta_1 \langle x_2, x_1 \rangle + \alpha_2 \beta_2 \langle x_2, x_2 \rangle + \cdots + \alpha_2 \beta_m \langle x_2, x_m \rangle \\ &\quad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ &\quad + \alpha_m \beta_1 \langle x_m, x_1 \rangle + \alpha_m \beta_2 \langle x_m, x_2 \rangle + \cdots + \alpha_m \beta_m \langle x_m, x_m \rangle\end{aligned}$$

This implies

$$\begin{aligned}\langle x, y \rangle &= [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_m] \begin{bmatrix} \langle x_1, x_1 \rangle & \langle x_1, x_2 \rangle & \cdots & \langle x_1, x_m \rangle \\ \langle x_2, x_1 \rangle & \langle x_2, x_2 \rangle & \cdots & \langle x_2, x_m \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle x_m, x_1 \rangle & \langle x_m, x_2 \rangle & \cdots & \langle x_m, x_m \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix} \\ &= [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_m] A \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}\end{aligned}$$

This implies

$$\begin{aligned}\langle x, y \rangle &= [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_m] \begin{bmatrix} \langle x_1, x_1 \rangle & \langle x_1, x_2 \rangle & \cdots & \langle x_1, x_m \rangle \\ \langle x_2, x_1 \rangle & \langle x_2, x_2 \rangle & \cdots & \langle x_2, x_m \rangle \\ \vdots & \vdots & \vdots & \vdots \\ \langle x_m, x_1 \rangle & \langle x_m, x_2 \rangle & \cdots & \langle x_m, x_m \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix} \\ &= [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_m] A \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}\end{aligned}$$

where $A = \begin{bmatrix} \langle x_1, x_1 \rangle & \langle x_1, x_2 \rangle & \cdots & \langle x_1, x_m \rangle \\ \langle x_2, x_1 \rangle & \langle x_2, x_2 \rangle & \cdots & \langle x_2, x_m \rangle \\ \vdots & \vdots & \vdots & \vdots \\ \langle x_m, x_1 \rangle & \langle x_m, x_2 \rangle & \cdots & \langle x_m, x_m \rangle \end{bmatrix} = [\langle x_i, x_j \rangle]$ is called the **matrix of inner product** with respect to ordered basis $\{x_1, x_2, \dots, x_m\}$.

Example

Consider the inner product space $(\mathbb{R}^3, \langle \rangle)$, where $\langle \rangle$ is given by

$$\left\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \right\rangle = x_1x_2 + y_1y_2 + z_1z_2.$$

Find the matrix of inner product space with respect to ordered basis $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

Example

Consider the inner product space $(\mathbb{R}^3, \langle \rangle)$, where $\langle \rangle$ is given by

$$\left\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \right\rangle = x_1 x_2 + y_1 y_2 + z_1 z_2.$$

Find the matrix of inner product space with respect to ordered basis

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Solution

The matrix of inner product space is given by

$$A = \begin{bmatrix} \langle x_1, x_1 \rangle & \langle x_1, x_2 \rangle & \langle x_1, x_3 \rangle \\ \langle x_2, x_1 \rangle & \langle x_2, x_2 \rangle & \langle x_2, x_3 \rangle \\ \langle x_3, x_1 \rangle & \langle x_3, x_2 \rangle & \langle x_3, x_3 \rangle \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Example

Consider the inner product space $(\mathbb{R}_2[x], \langle \rangle)$, where $\langle \rangle$ is given by

$$\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx.$$

Find the matrix of inner product space with respect to the ordered basis $\{1, x, x^2\}$.

Solution

The matrix of given inner product is given by

$$A = \begin{bmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle & \langle 1, x^2 \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle & \langle x, x^2 \rangle \\ \langle x^2, 1 \rangle & \langle x^2, x \rangle & \langle x^2, x^2 \rangle \end{bmatrix}$$

$$\langle 1, 1 \rangle = \int_0^1 1 \cdot 1 dx = 1 \qquad \langle 1, x \rangle = \int_0^1 1 \cdot x dx = \frac{1}{2} \qquad \langle 1, x^2 \rangle = \int_0^1 1 \cdot x^2 dx = \frac{1}{3}$$

$$\langle x, 1 \rangle = \int_0^1 x \cdot 1 dx = \frac{1}{2} \qquad \langle x, x \rangle = \int_0^1 x \cdot x dx = \frac{1}{3} \qquad \langle x, x^2 \rangle = \int_0^1 x \cdot x^2 dx = \frac{1}{4}$$

$$\langle x^2, 1 \rangle = \int_0^1 x^2 \cdot 1 dx = \frac{1}{3} \qquad \langle x^2, x \rangle = \int_0^1 x^2 \cdot x dx = \frac{1}{4} \qquad \langle x^2, x^2 \rangle = \int_0^1 x^2 \cdot x^2 dx = \frac{1}{5}$$

Thus,

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$

Exercise 9:

The vectors $u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $u_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ form a basis S for Euclidean space $\mathbb{R}^3$. Find the matrix A that represents the inner product in $\mathbb{R}^3$ relative to this basis S .

Orthonormal Vectors

Let $(V, \langle \rangle)$ be an inner product space. Then a set of vectors $\{x_1, x_2, \dots, x_k\}$ in V is said to be orthonormal if $\|x_i\| = 1$ and $\langle x_i, x_j \rangle = 0$ for $i \neq j$.

A subset $\{x_1, x_2, \dots, x_k\}$ of V is said to be an orthonormal basis for V if it is a basis of V and is orthonormal set.

Example

Consider the inner product space $(\mathbb{R}^2, \langle \cdot, \cdot \rangle)$, where

$$\left\langle \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right\rangle = x_1 x_2 + y_1 y_2.$$

Then $\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}$ is an orthonormal basis of $\mathbb{R}^2$ as

$$\left\| \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\| = \left\| \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\| = 1$$

and

$$\left\langle \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\rangle = 0$$

Exercise 10:

Show that the set $S = \left\{ \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}, \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix}, \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ -\frac{1}{3} \end{bmatrix} \right\}$ is an orthonormal set of vectors in $\mathbb{R}^3$ with Euclidean inner product.

Exercise 11:

If $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \right\}$ is an orthogonal set of vectors in $\mathbb{R}^3$ then find an orthonormal set.

Result

If $\mathcal{B} = \{v_1, v_2, \dots, v_m\}$ is an orthonormal basis of an inner product space V and x is any vector in V then

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \cdots + \langle x, v_m \rangle v_m.$$

Result

If $\mathcal{B} = \{v_1, v_2, \dots, v_m\}$ is an orthonormal basis of an inner product space V and x is any vector in V then

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \cdots + \langle x, v_m \rangle v_m.$$

Proof

Since $\mathcal{B}$ is a basis of V and $x \in V$ and hence let

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_m v_m.$$

However, for each $i = 1, 2, \dots, m$, we have

$$\begin{aligned}\langle x, v_i \rangle &= \langle \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_i v_i + \cdots + \alpha_m v_m, v_i \rangle \\ &= \alpha_1 \langle v_1, v_i \rangle + \alpha_2 \langle v_2, v_i \rangle + \cdots + \alpha_i \langle v_i, v_i \rangle + \cdots + \alpha_m \langle v_m, v_i \rangle \\ &= \alpha_i \|v_i\|^2 = \alpha_i\end{aligned}$$

Therefore

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \cdots + \langle x, v_m \rangle v_m.$$

Orthogonal Projection

Let U and W be subspaces of an inner product space $(V, \langle \cdot, \cdot \rangle)$.

- ❁ Two subspaces U and W are said to be orthogonal, written by $U \perp W$, if $\langle u, w \rangle = 0$ for all $u \in U$ and $w \in W$.
- ❁ The set of all vectors in V that are orthogonal to every vector in U is called the orthogonal complement of U and is denoted by $U^\perp$, i.e.,

$$U^\perp = \{v \in V : \langle u, v \rangle = 0 \ \forall u \in U\}$$

Note

- ① The orthogonal complement $U^\perp$ of U is a subspace of the vector space V .
- ② $U \cap U^\perp = \{0\}$ and $V = U + U^\perp$.
- ③ $\dim V = \dim U + \dim U^\perp$.

Definition (Direct Sum)

Let V be a vector space and let U and W be two subspaces of V such that

$$V = U + W \quad \text{and} \quad U \cap W = \{0\}.$$

Then V is called direct sum of U and W and is written as $V = U \oplus W$.

Example

Let $V = \mathbb{R}^2$ and let U and W be two subspaces of $\mathbb{R}^2$ given by

$$U = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} : x \in \mathbb{R} \right\}$$

$$W = \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} : y \in \mathbb{R} \right\}$$

Then show that $\mathbb{R}^2 = U \oplus W$.

Solution

Suppose $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ then

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \end{bmatrix}$$

Now $\begin{bmatrix} x \\ 0 \end{bmatrix} \in U$ and $\begin{bmatrix} 0 \\ y \end{bmatrix} \in W$ and hence $V = U + W$. Next, suppose $\begin{bmatrix} a \\ b \end{bmatrix} \in U \cap W$.

Then

$$\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} \in U \quad \text{and} \quad \begin{bmatrix} a \\ b \end{bmatrix} \in W$$

$$\Rightarrow b = 0 \quad \text{and} \quad a = 0$$

$$\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus $U \cap W = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$. This shows that $\mathbb{R}^2$ is direct sum of U and W , i.e.,

$$\mathbb{R}^2 = U \oplus W$$

Example

Describe the orthogonal complement $V^\perp$ of V where

$$V = \text{span} \left(\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right),$$

where $\langle \rangle$ is Euclidean inner product.

Solution

The subspace V of $\mathbb{R}^3$ is spanned by the two vectors $v_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

Therefore its orthogonal complement $V^\perp$ is the set of vectors which are orthogonal to both v_1 and v_2 , i.e.,

$$V^\perp = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \left\langle \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \right\rangle = 0 \text{ and } \left\langle \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle = 0 \right\}.$$

Let $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in V^\perp$ then

$$\left\langle \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \right\rangle = 0 \implies x - 3y + 2z = 0 \quad (3)$$

$$\left\langle \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle = 0 \implies y + z = 0 \quad (4)$$

The augmented matrix of above homogeneous system of linear equation is

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \xrightarrow{RREF} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right].$$

System corresponding to above RREF is

$$x + 5z = 0 \implies x = -5z$$

$$y + z = 0 \implies y = -z$$

Thus,

$$V^\perp = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x = -5z \text{ \& } y = -z \right\} = \left\{ z \begin{bmatrix} -5 \\ -1 \\ 1 \end{bmatrix} : z \in \mathbb{R} \right\}.$$

Exercise 12:

Compute the orthogonal complement of the following subspace of $\mathbb{R}^3$

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : 3x + 2y = z \right\},$$

where $\langle \rangle$ is Euclidean inner product.

Exercise 13:

Compute $W^\perp$, where

$$W = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right),$$

with respect to Euclidean inner product.

Exercise 14:

Find all vectors perpendicular to $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ with respect to Euclidean inner product.

Exercise 15:

Find a basis for the subspace u of $\mathbb{R}^3$, where $u = \text{span} \left(\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \right)$.

Definition(Orthogonal Projection)

Let $(V, \langle \rangle)$ be an inner product space, and let U be a subspace of V so that $V = U \oplus U^\perp$. Then the projection of V onto U along $U^\perp$ is called the **orthogonal projection** of V onto U , denoted Proj_U . For $x \in V$, the component vector $\text{Proj}_U(x) \in U$ is called the **orthogonal projection** of x into U .

Result

Let U be a subspace of an inner product space $(V, \langle \rangle)$, and let $\{u_1, u_2, \dots, u_m\}$ be an orthogonal basis for U . Then, for any $x \in V$, the orthogonal projection $\text{Proj}_U(x)$ of x into U is

$$\text{Proj}_U(x) = \frac{\langle x, u_1 \rangle}{\|u_1\|^2} u_1 + \frac{\langle x, u_2 \rangle}{\|u_2\|^2} u_2 + \cdots + \frac{\langle x, u_m \rangle}{\|u_m\|^2} u_m.$$

Example

Consider the inner product space $(\mathbb{R}^3, \langle \cdot \rangle)$, where $\langle \cdot \rangle$ is Euclidean inner product. Let W be the subspace of $\mathbb{R}^3$ given by

$$W = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right).$$

Find the projection of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto W .

Solution

Since vectors $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ are orthogonal vectors and hence the orthogonal projection of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto W is given by

$$\begin{aligned}\text{Proj}_U \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) &= \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\| \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\|^2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\rangle}{\left\| \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\|^2} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \\ &= \frac{2+3}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{-2+3}{2} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}\end{aligned}$$

Example

Let $V = \mathbb{R}_3[x]$, the vector space of polynomials of degree less than or equal to 3 equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx \quad \text{for any } f, g \in \mathbb{R}_3[x].$$

Let W be the subspace of V given by $W = \text{span}(1, x)$. Let $f(x) = x^2$. Find the orthogonal projection $\text{Proj}_W(f)$ of f onto W .

The Gram-Schmidt orthogonalization process

The Gram-Schmidt orthogonalization process

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let $\{x_1, x_2, \dots, x_m\}$ be a basis for V . Let

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$v_3 = x_3 - \frac{\langle x_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle x_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots$$

$$v_m = x_m - \frac{\langle x_m, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle x_m, v_2 \rangle}{\|v_2\|^2} v_2 - \dots - \frac{\langle x_m, v_{m-1} \rangle}{\|v_{m-1}\|^2} v_{m-1}.$$

Then $\{v_1, v_2, \dots, v_m\}$ is an orthogonal basis of V .

Note

✿ If $\{v_1, v_2, \dots, v_m\}$ is an orthogonal basis of V then $\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \dots, \frac{v_m}{\|v_m\|} \right\}$ will be an orthonormal basis of V .

Note

- ✿ If $\{v_1, v_2, \dots, v_m\}$ is an orthogonal basis of V then $\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \dots, \frac{v_m}{\|v_m\|} \right\}$ will be an orthonormal basis of V .
- ✿ Any basis of an inner product space $(V, \langle \rangle)$ is either an orthonormal basis or can be converted into orthonormal basis by using the Gram-Schmidt orthogonalization process.

Example

Consider the inner product space $(\mathbb{R}^2, \langle \cdot \rangle)$ where $\langle \cdot \rangle$ is Euclidean inner product. Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ be a basis of $\mathbb{R}^2$. Use the Gram-Schmidt orthogonalization process to find an orthogonal basis.

Solution

Setting $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Then using Gram-Schmidt orthogonalization process, we get

$$\begin{aligned} v_1 &= x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ v_2 &= x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1 \\ &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{3}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \end{aligned}$$

Thus $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \right\}$ is an orthogonal basis of $\mathbb{R}^2$.

Example

Consider the inner product space $(\mathbb{R}_2[x], \langle \cdot \rangle)$, equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$$

Let $\mathcal{B} = \{1, x, x^2\}$ be a basis of $\mathbb{R}_2[x]$. Find an orthogonal basis of $\mathbb{R}_2[x]$.

Solution

Setting $f_1 = 1$, $f_2 = x$, $f_3 = x^2$. Then

$$v_1 = f_1 = 1$$

$$\begin{aligned} v_2 &= f_2 - \frac{\langle f_2, v_1 \rangle}{\|v_1\|^2} v_1 \\ &= x - \frac{\langle x, 1 \rangle}{\|1\|^2} 1 = x - \frac{1}{2} \end{aligned}$$

$$\begin{aligned} v_3 &= f_3 - \frac{\langle f_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle f_3, v_2 \rangle}{\|v_2\|^2} v_2 \\ &= x^2 - x + \frac{1}{6} \end{aligned}$$

Thus $\{1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}\}$ is an orthogonal basis of $\mathbb{R}_2[x]$.

Example

Find the closest point to x in the subspace W of $\mathbb{R}^4$ spanned by v_1 and v_2 , where

$$x = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 1 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

Solution

Since $\langle v_1, v_2 \rangle = 3 \cdot 1 + 1 \cdot (-1) + (-1) \cdot 1 + 1 \cdot (-1) = 0$ and $\{v_1, v_2\}$ is linearly independent and hence $\{v_1, v_2\}$ forms an orthogonal basis of W . The closest point to x in the subspace W is $\text{Proj}_W(x)$. Thus

$$\begin{aligned}\text{Proj}_W(x) &= \frac{\langle x, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle x, v_2 \rangle}{\|v_2\|^2} v_2 \\&= \frac{9 + 1 - 5 + 1}{9 + 1 + 1 + 1} \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix} + \frac{3 - 1 + 5 - 1}{1 + 1 + 1 + 1} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \\&= \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}\end{aligned}$$

Example

Find an orthogonal basis for the column space of the following matrix

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

Solution

First of all let us find a basis of $\text{col}(A)$

$$\begin{array}{l} \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - R_1 \\ R_5 \rightarrow R_5 - R_1 \end{array} \end{array} \begin{array}{l} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & -1 \\ 0 & 6 & 2 \\ 0 & -6 & 2 \\ 0 & 0 & 4 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 + 2R_2 \end{array} \end{array} \begin{array}{l} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 4 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_5 \rightarrow R_5 - R_4 \end{array} \begin{array}{l} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \end{array} R_3 \leftrightarrow \begin{array}{l} \begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Each column contains a leading entry and hence a basis for $\text{Col}(A)$ is

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \\ -4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ -4 \\ -3 \\ 7 \\ 7 \end{bmatrix} \right\}$$

Now, we convert this basis into orthogonal basis using Gram-Schmidt orthogonal-

ization process. Setting $x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 2 \\ 1 \\ 4 \\ -4 \\ 2 \end{bmatrix}$, $x_3 = \begin{bmatrix} 5 \\ -4 \\ -3 \\ 7 \\ 7 \end{bmatrix}$. Then

$$v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}$$

$$v_3 = x_3 - \frac{\langle x_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle x_3, v_2 \rangle}{\|v_2\|^2} v_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}$$

$\{v_1, v_2, v_3\}$ is an orthogonal basis.

Example

Use the Gram-Schmidt orthogonalization process on the Euclidean space $\mathbb{R}^4$ to transform the basis

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

into an orthonormal basis.

Solution

Setting $x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $x_3 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$, $x_4 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$. Then

$$v_1 = x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{bmatrix} -1 \\ \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

$$v_3 = x_3 - \frac{\langle x_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle x_3, v_2 \rangle}{\|v_2\|^2} v_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$v_4 = x_4 - \frac{\langle x_4, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle x_4, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle x_4, v_3 \rangle}{\|v_3\|^2} v_3 = \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ -1 \end{bmatrix}$$

Thus

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}, \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{3} \\ -1 \end{bmatrix} \right\}$$

is an orthogonal basis of $\mathbb{R}^4$ and dividing each vector by its magnitude, we get

$$\left\{ \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}, \begin{bmatrix} -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \right\}$$

which is an orthonormal basis of $\mathbb{R}^4$.

Example

Find the point on the plane $x - y - z = 0$ that is closest to $p = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$.

Solution

Given plane can be expressed as subspace of $\mathbb{R}^3$ as following

$$\begin{aligned} W &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x - y - z = 0 \right\} = \left\{ \begin{bmatrix} y + z \\ y \\ z \end{bmatrix} : y, z \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} z \\ 0 \\ z \end{bmatrix} : y, z \in \mathbb{R} \right\} = \left\{ y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} : y, z \in \mathbb{R} \right\} \\ &= \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) \end{aligned}$$

Since $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a linearly independent subset of $\mathbb{R}^3$ and hence this forms a basis of W .

Next, we'll find an orthogonal basis using Gram-Schmidt orthogonalization process.

Setting $x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, we have

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

Thus, $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \right\}$ is an orthogonal basis of W . Therefore the point on the

plane $x - y - z = 0$ that is closest to $p = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ is

$$\text{Proj}_W(p) = \frac{\langle p, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle p, v_2 \rangle}{\|v_2\|^2} v_2 = \begin{bmatrix} \frac{4}{3} \\ \frac{5}{3} \\ -\frac{1}{3} \end{bmatrix}$$

Exercise 16:

Construct an orthogonal basis for $\mathbb{R}_2[x]$ with respect to the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$$

by applying the Gram-Schmidt process to the basis $\{1, x, x^2\}$.

Exercise 17:

Find an orthonormal basis for the solution space of the homogeneous system of linear equations

$$\begin{aligned}x_1 + x_2 + 7x_4 &= 0 \\ 2x_1 + x_2 + 2x_3 + 6x_4 &= 0\end{aligned}$$

Exercise 18:

Apply Gram-Schmidt process to obtain an orthonormal basis of $\mathbb{R}^3$ over $\mathbb{R}$ from the basis

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

using Euclidean inner product.

Exercise 19:

Apply Gram-Schmidt process to obtain an orthonormal basis of $\mathbb{R}^3$ over $\mathbb{R}$ from the basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} \right\}$$

using Euclidean inner product.